Approaches to Image Binarization in Current Automated Fingerprint Identification Systems

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Abstract* – Some of the current approaches to fingerprint image binarization are investigated. The importance of the binarization phase of processing as it relates to the identification of accurate minutiae points is established. An overview of the most commonly used approaches in this area is then presented. The strengths and weaknesses of these approaches are explored and suggestions for future research and improvements are set forth.

I. INTRODUCTION

The use of biometric information to identify individuals has become a valuable security and forensic tool in recent years [1]. Of the biometric features available, the fingerprint continues to be one of the most commonly used. This is, in part, due to its technological familiarity, ease of collection, and public acceptance [2],[3]. As a result, the development of faster and more accurate automated fingerprint identification systems (AFIS) continues to be an important research area. In a typical AFIS, two fingerprints are matched by comparing their sets of local features, or minutiae. While these features can be classified into a diverse set of shapes, the two most commonly found are ridge endings and bifurcations. An example of these two local features can be seen in Figure 1.

Since ridge endings and bifurcations are the most common features, most AFIS use the locations, angles, and types of these two features when testing for a match. The quality of the AFIS is therefore heavily dependent on the system's ability to accurately extract the locations of these points. As a result, most AFIS will pass a fingerprint image through several steps of processing in order to increase the reliability of the information extracted. Figure 2 shows a diagram of some of these stages. Of these processing stages, one of the most critical is binarization. The binarization stage is responsible for converting the grayscale fingerprint image into a black and white image that can be thinned to single pixel width for easy minutiae extraction. If a poor binarization algorithm is used, an accurate skeleton cannot be obtained and the minutiae extraction will not provide an accurate representation of the fingerprint image. On the other hand, if a highly accurate binary image is produced, the skeletonization process will go smoothly and a minimum of invalid minutiae points will be introduced, thus maximizing the ability to accurately match the fingerprint. In the remainder of this paper, the importance of binarization will be explored and some of the common methods that are currently used to perform this critical step in minutiae extraction will be examined. Section 2 will explain the critical nature of the binarization step and look at some common errors that occur. Section 3 will examine some of the current binarization methods, and section 4 will draw some conclusions and set forth some areas for improvement and future research.

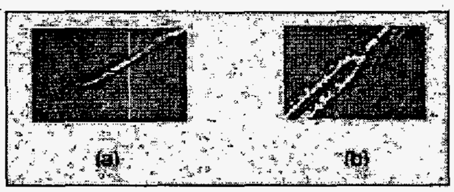


Figure 1. An example of the two most common ridge features: (a) Ridge Endings, and (b) Ridge Bifurcations

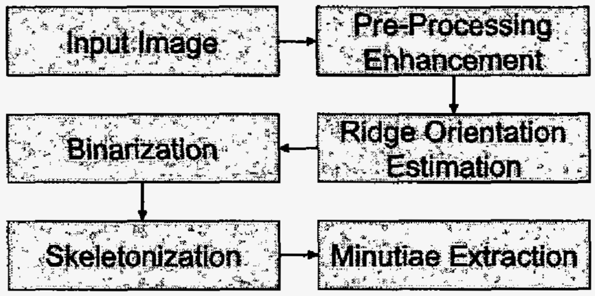


Figure 2. An example of the common processing stages used in minutiae extraction.

II. BINARIZATION

Of all the steps outlined in Figure 2, binarization is one of the most critical. Even with the best extraction and matching methods, a poor binarization algorithm can destroy the accuracy and reliability of an AFIS. For example, Figure 3 illustrates the importance of a good binarization algorithm. The figure displays an original fingerprint image along with the results of a poor binarization approach. The resulting

(a) (b) (c)

Figure 3. An example illustrating the importance of a good binarization step in an AFIS. In this example, the original image (a) was binarized with a poor binarization method to achieve (b). The poor quality of the binarization step is further illustrated by examining the errors introduced into the thinned image (c).

skeleton is also provided so that it is easier to see how the inadequate binarization step can impact future extraction stages.

In the best possible case, a good binarization algorithm should be capable of providing an image consisting of smooth, flowing, black ridges of uniform width on a white background. An image of this type can easily be transformed into a thinned, image with a minimum of false points. When the binarized image deviates from this ideal case, however, problems begin to develop. There are four basic flaws in a binarized image that can lead to problems in the thinning stage. These are:

1) Breaks in the ridge flow,
2) Ridges that are merged together,
3) Rough edges on the ridge lines, and
4) Holes in the middle of ridge lines.

Any of these binarization errors can lead to the development of structures in the thinned image that will be incorrectly identified as minutiae points. Figure 4 illustrates these four problems and the resulting false minutiae structures that appear in the thinned images. As the figure shows, the binarization processing stage is critical to proper minutiae extraction.

III. BINARIZATION APPROACHES

Now, that the critical nature of binarization has been established, some of the most commonly used binarization algorithms can be examined. Of the multitude of methods that have been used in fingerprint identification systems, most of them fall into one of three categories:

1) Global Approaches,
2) Neighborhood-Based Approaches, and
3) Filter-Based Approaches.

The first category that will be examined is that of global approaches. A global binarization approach, as the name suggests, is an approach that is applied to the image as a whole. The one common binarization approach that falls into this category is global thresholding. In this method, a threshold is determined for a given image, typically based upon some statistical measure of the image, and then all pixels with values above the threshold become white and those with values below become black [4]. This is one of the oldest and simplest approaches, and as such, it usually does not do a very good job.

While its simplicity tends to make it a very fast algorithm (only one comparison is needed per image pixel), global thresholding suffers from its lack of flexibility. If the image contains areas of varying intensity, such as smudged or lightened areas, a global approach will tend to fail for these

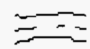
Description	Binarization Error	Resulting Skeleton Error
Joined Ridges		
Ridge Flow Breaks		
Rough Ridge Edges		
Holes in Ridges		

Figure 4. An illustration of some of the common errors that can occur in binarization and the errors they produce in the image skeleton.

Figure 5. An example of local adaptive thresholding used for fingerprint binarization. The binarized image (b) quality is clearly better then that obtained by global thresholding in figure 3, but there are still many flaws evident in the skeleton image (c).

areas and leave large areas of the fingerprint unusable. No reliable minutiae can be extracted in these areas and the resulting large blobs tend to leave artifacts in the skeleton. The binarization method used in Figure 3 is a global thresholding approach. Since the use of global methods shows obvious disadvantages from their lack of local adaptability, the second category of methods became necessary.

The second category of approaches includes methods that base their outcomes on local neighborhoods within the image. In this area there are three common approaches in use. These are:

1) Local Adaptive Thresholding,
2) Oriented Grid, and
3) Peak Detection.

Each of these approaches has its own strengths and weaknesses.

The most logical extension of the global thresholding approach is to use the same basic idea, but to limit it to local neighborhoods within the image. A typical method of this type will divided the image into a series of blocks. Each of these blocks is then converted into a binary representation using methods similar to those used in global thresholding [4]. This approach maintains the speed advantage that was one of the major benefits of global thresholding while reducing the negative impact of uneven lighting conditions within the image. Thus, adaptive local thresholding handles images with smudged and lightened areas better than global thresholding can. While this does improve the overall binarized image, the resulting image still suffers from many of the common binarization defects that lead to false minutiae detection. An example of adaptive local thresholding can be found in Figure 5. It can easily be seen in this figure that the resulting skeleton contains fewer false structures, but this method of binarization can do nothing to separate joined ridges, join broken ridges, smooth ridge edges or fill the holes in the ridges. It simply provides a cleaner and more accurate binarization than the global approach.

The second approach that utilizes local pixel neighborhoods is the oriented grid approach. This approach, which is utilized in the NIST minutiae extraction algorithm, centers an m x n grid (NIST uses a 7 x 9 grid) on each pixel in the image [5]. This grid is oriented in the direction of the detected ridge flow associated with the center pixel. This arrangement is illustrated in Figure 6. The intensities of the pixels located along each of the rotated grid rows are then summed. The binary value assigned to the center pixel is then determined by comparing the sum of the center row multiplied by the number of rows to the sum of the entire grid. In other words if:

$$nr\sum_{k=1}^{nc} G(cr,k) < \sum_{i=1}^{nr}\sum_{j=1}^{nc} G(i,j) \qquad (1)$$

then the center pixel is set to black (0). In this equation, *nr* is the number of rows in the grid, *nc* is the number of columns, *cr* is the index of the center row, and *G(i,j)* represents the pixel intensity at the i^{th} row and j^{th} column of the rotated grid. If the above condition is not true, the center pixel is set to

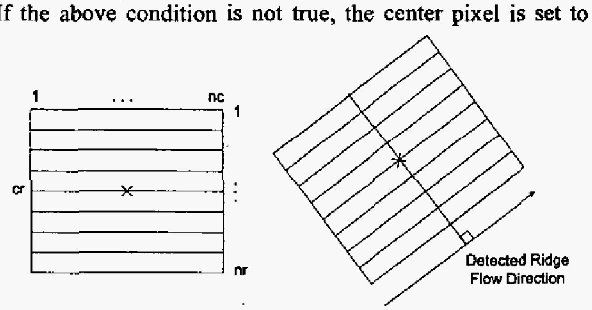


Figure 6. An illustration of the oriented grid approach. The grid used to determine whether the given center pixel is black or white is oriented along the detected ridge flow direction associated with the center pixel [4].

Figure 7. An example of the oriented grid approach used for fingerprint binarization. The binarized image (b) quality is clearly better then that obtained by either of the thresholding approaches in figures 3 and 5. The skeleton image (c) contains fewer bridges between ridges, but still contains ridge spurs, breaks, and loops caused by holes.

white (1).

This type of approach suffers in the area of speed. Since sums must be taken along the rows of the local oriented neighborhoods for each pixel in the image, there are considerably more computational steps involved than with plain thresholding. In addition, this method can do nothing to separate joined ridges, connect large ridge breaks, fill in large holes in the ridge, or provide smooth ridge edges. On the positive side, however, this approach produces a much higher quality binary image than either global or adaptive thresholding. Since it looks at the pixels along the ridge, it is capable of filling in small holes and connecting short breaks in the ridge structure. Figure 7 illustrates an example of this approach. The image clearly shows an improvement in quality over the thresholding approaches, however, there are still several errors in the skeleton image. Examples of several problems caused by broken ridges, rough edges and joined ridges are clearly evident. The most evident problem is the small loop structures in the ridge flow of the skeleton image. These are caused by holes in the binary ridges. It should be noted that employing a hole filling algorithm after binarization could improve the image quality at the expense of additional processing time.

The final neighborhood approach is a method known as peak detection. This method, first proposed by Ratha, Chen, and Jain [6], creates a 16 x 16 oriented neighborhood around each pixel in the image. The pixel intensities are then projected onto the center portion of the window to form an orthogonal ridge profile. Once the profile is obtained, it is smoothed using a local averaging approach. The peaks are then detected in the profile. The peak pixel and its two neighboring pixels to each side are kept as ridge pixels and the rest are set as background [6]. This concept is illustrated in Figure 8.

This approach also suffers in the area of speed when compared to simple thresholding approaches, due to the fact that an entire 16 x 16 neighborhood must be processed for each point in the image. On the positive side, this approach can partially adapt to breaks in ridges and holes in the ridge flow. In addition, since this method only keeps two values to each side of the peaks, it is capable of separating lightly connected ridges and the ridge edges tend to be smoother than the previos methods. While it is a slightly different approach, this method tends to produce output that is similar in quality to the oriented grid approach. However, since it keeps fewer pixels, just two to each side of the peak, it tends to skeletonize more quickly.

The final category of binarization approaches are filter-based approaches. These methods, as the name suggests, all rely on the application of a filter to the grayscale fingerprint image that either directly produces a binary result or that produces an output that is easily converted to binary by simple thresholding. These filters may be applied in the spatial or frequency domains and are usually filters that are tuned to the frequency and orientation of the ridges within a praticular portion of the image. Approaches in this area have provided some of the most promising results in the area of fingerprint image binarization. As a result, variations on this approach have been proposed by a large number of researchers [7 – 13]. One common example of this approach, originally proposed by Hong, Wan, and Jain, makes use of an enhancement phase utilizing even symmetric Gabor filters.

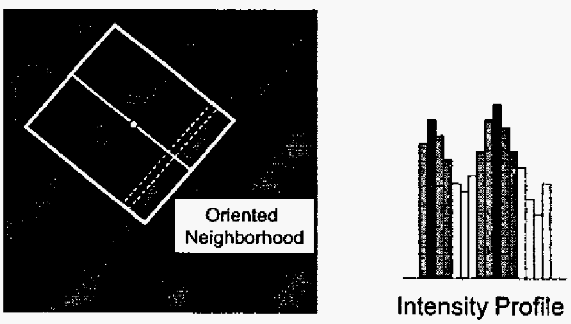


Figure 8. An illustration of the peak detection method. The intensities in the neighborhood are projected onto the orthogonal center section to obtain an intensity profile

Figure 10. An example of the results obtained by applying a filter-based binarization to a fingerprint image. This particular approach makes use of even symmetric Gabor filters that are tuned to the characteristics of a local neighborhood. Of the methods proposed in this paper, this approach provides the highest quality results.

The output of this enhancement phase is easily thresholded to provide a binary image. In this approach, an even symmetric Gabor filter is oriented along the local ridge direction and tuned to the local frequency for a given neighborhood. After performing the required filtering, a final binary image is constructed from each of the enhanced neighborhoods. Figure 9 illustrates a typical Gabor filter, and Figure 10 demonstrates the results of binarizing an image using this approach. As the figure clearly demonstrates, this approach can provide a very high quality image. Unfortunately, it is also the most computationally intensive of all the methods discussed in this paper. Aside from its speed, however, it does a good job of separating joined ridges, bridging small to moderate ridge breaks, closing holes in the ridge flow, and providing smooth edges. While it can provide good results, these results rely heavily on a number of factors. First, it is heavily dependent on the orientation map of the fingerprint. If an accurate orientation map cannot be obtained, this method will provide unpredictable results. Second, the filter parameters must be tuned properly. The standard deviations, frequency, and orientation must all be set properly for a given region or the results can be unsatisfactory. For example, if the frequency is set too high, the filter tends to create new ridges by splitting existing ones. On the other hand if the frequency is too low, joined ridges may not be separated, or closely spaced true ridges may be merged. Thus, these methods do provide excellent results, but can introduce false structures into the image if not tuned properly. There have been recent attempts to modify this Gabor filter approach to make it more robust and to provide better results [14].

$$G(x,y;f,\theta)=\exp\left\{-\frac{1}{2}\left[\frac{x'^2}{\delta_{x'}^2}+\frac{y'^2}{\delta_{y'}^2}\right]\right\}\cos(2\pi f x')$$

$$y' = x\cos(\theta) - y\sin(\theta)$$
$$x' = x\sin(\theta) + y\cos(\theta)$$

f *(Spatial Frequency)*
$\delta_{x'}$, $\delta_{y'}$ *(Std. Deviations)*
θ *(Filter Angle)*

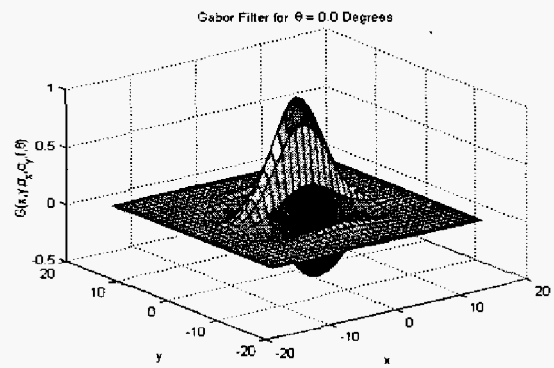


Figure 9. An illustration of the spatial equation for an even symmetric Gabor filter and the plot of a typical Gabor filter oriented at 0 degrees.

IV. CONCLUSIONS

In this paper, some of the common methods used to binarize fingerprint images have been examined. These approaches fall into several categories and take a number of different approached to achieve their final goal, but they all strive to provide the best possible binary image to subsequent stages of the automated fingerprint recognition system. Each of the methods reviewed has its own set of strengths and weaknesses. Table 1 helps to outline these strengths and weaknesses by organizing the different approaches and ranking their abilities in the areas of execution speed, algorithm complexity, output quality, and method robustness.

As the table demonstrates, the approaches with high speed tend to produce poor results and do not adapt well to a variety of images, particularly low quality fingerprint images. The algorithms that look at local neighborhoods take a bit

Table 1. An evaluation of the systems reviewed in this paper with respect to complexity, speed, quality of output, and algorithm robustness.

Algorithm	Complexity	Speed	Quality	Robustness
Global Thresholding	Low	Very Fast	Poor	Low
Local Adaptive Thresholding	Low	Fast	Fair	Fair
Oriented Grid	Medium	Medium	Good	Good
Peak Detection	Medium	Medium	Good	Good
Filter-Based	High	Slow	Excellent	Good

longer to process, but overall tend to produce better binary images and are capable of adapting to lower quality images. The filter-based approaches, while currently the slowest of the approaches do produce the best results and tend to be fairly robust to changes in the input images. They are capable of producing good results even in the presence of low quality input images. They are, however, not a perfect solution. There is a great deal of research yet to be done. This research currently seems to be centered around the newest binarization category, filter-based approaches. Future research directions in this area can be divided into two areas, enhancing speed and improving output quality and robustness. In the area of improved speed, research will involve the development of better and faster filters that can accomplish comparable results to current filter-based methods. In the area of quality and robustness, work must be done to develop filters that can provide the benefits of the current designs while minimizing their negative aspects, such as the introduction of false ridges and structures. This work could also include improvements in current filter designs to make them more robust to different types of input images. Thus, while there currently exist several promising methods of fingerprint image binarization there is still a great deal of room for improvement and future work. As a result, fingerprint image binarization will remain a critical step in the fingerprint detail extraction process and an improtant area for future research.

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